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1 January 2024

SUPERSEDING
DOD-STD-1399 (NAVY) SECTION 441
1 June 1982

DEPARTMENT OF DEFENSE

INTERFACE STANDARD FOR SHIPBOARD SYSTEMS

SECTION 441

PRECISE TIME AND TIME INTERVAL

(PTTI)

AMSC N/A **FSC 1990**

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2. Recommended corrections, additions, or deletions should be addressed to Naval Information Warfare (NAVWAR) Systems Command, and (Code 5.0) San Diego, CA 92110.

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FOREWORD

Purpose. This section defines the standard interface requirement for, and the constraints on, the design of shipboard equipment which will utilize the Precise Time and Time Interval (PTTI) information distributed via the platform Positioning, Navigation and Timing (PNT) Systems. The PTTI reference for surface ships will be the Global Positioning System (GPS) based PNT Service (GPNTS) and in some cases may provide direct distribution of PTTI signals. In other cases the distribution may be handled by other systems disciplined by GPNTS such as the Naval Critical Distribution System (NCDS), Time Distribution Units (TDUs), or the Integrated Radio Communications System (iRCS). For the submarine force the distribution will be handled by the Time and Frequency Distribution System (TFDS).

Nature of the interface. The Navy's PTTI program is an effort to disseminate highly accurate PTTI reference worldwide from the U.S. Naval Observatory (USNO) to Navy ships and other platforms and installations having a need for such references. These references are distributed within the ship via GPNTS and other systems. It is essential that the interface between the PTTI GPNTS and other systems outputs and the various user equipment's be engineered so that the stability and accuracy of the outputs are not degraded and that the user equipment's may accept and process the reference outputs effectively.

Structure. The technical content of this section first delineates the interface characteristics in terms of available frequency and time parameters, impedance, voltage, capacities, and so forth. The constraints on equipment design and installation necessary to achieve shipboard compatibility with these characteristics are then established.

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FIGURES

Figure
1 Interface
2 PTTI system concept
3 PTTI dissemination subsystem concept
4 PTTI platform distribution system concept

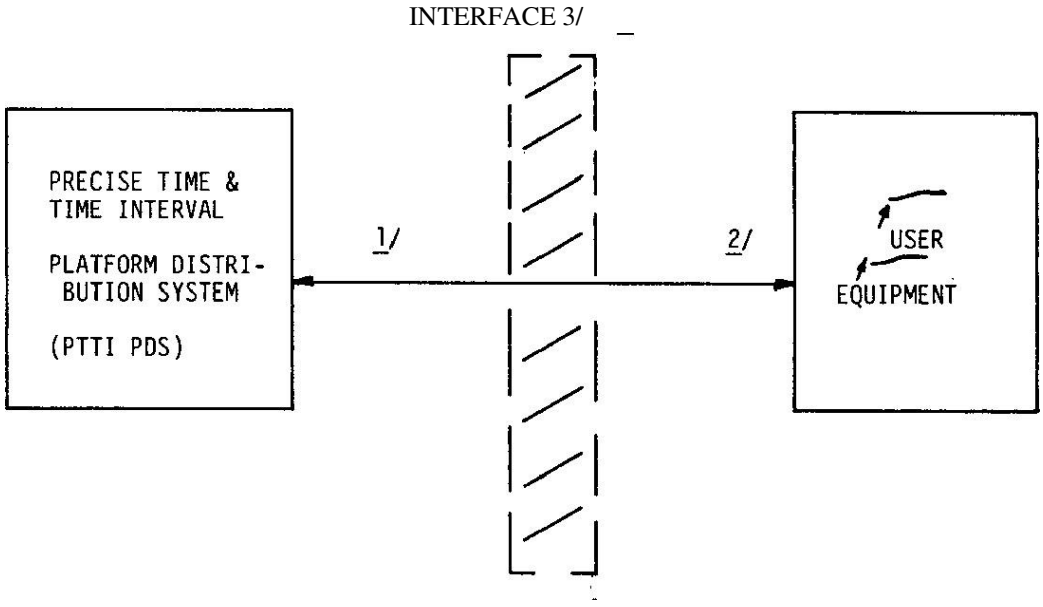
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1. SCOPE

1.1 General. Policies and procedures established by MIL-STD-1399 are mandatory. This section and the basic standard (that is, DoD-STD-1399) are to be viewed as an integral single document.

1.2 Scope. This section establishes interface requirements for shipboard equipment utilizing the PTTI Platform Distribution System (PDS) in order to ensure compatibility between the user equipment and the PTTI PDS.

1.3 Interface. The basic characteristics and constraint categories concerned with this interface are shown symbolically in FIGURE 1 (see section 3 "Definitions" of MIL-STD-1399):



BASIC CATEGORIES

1/ CHARACTERISTICS

Precise time
Precise time interval (frequency)
Output voltage
Capacity

2/ CONSTRAINTS

Compatibility

3/ The interface is located at the input connector Of the user equipment.

FIGURE l. Interface.

The particular interface characteristics and constraints pertinent to this section are described in 5.2 and 5.3

1.4 Applicability. The criteria of this section are applicable to new ship acquisitions, ship modernizations, and ship conversions wherein a requirement exists for equipment utilizing the outputs of the PTTI PDS. They apply also to ships having equipment to be interfaced in part with the PTTI PDS to the extent indicated (for example, frequency support only).

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

ICD-GPS-060	GPS Have Quick time reference
ICD-GPS-060B PTTI	GPS User Equipment (Phase III) Interface Control Document

(Copies can be found at ICD-GPS-060B.pdf (uscg.gov).

GPNTS TRD version 3.1	Global Positioning System (GPS)-based Positioning, Navigation, and Timing Service (GPNTS)

(Copies can be requested at PMW/A 170 Global Positioning System (GPS) Based Positioning, Navigation and Timing (PNT) Service (GPNTS) Production SPAWAR_Headquarters_MKTSVY_104712 (govtribe.com))

IRIG 200-16	IRIG Serial Time Code Formats, August 2016

(Copies can be found at 200-16_IRIG_Serial_Time_Code_Formats.pdf (menlosecurity.com))

IETF-RFC-5905	Network Time Protocol Version 4

(Copies can be found at RFC 5905 - Network Time Protocol Version 4: Protocol and Algorithms Specification (ietf.org)

IEEE-1588G-2022	IEEE Standard for a Precision Clock Synchronization Protocol

(Copies can be found at IEEE SA - Search)

STANDARDS

SYSCOMTA-ARC-023-PNTRA-V5.0	Cybersecurity Technical Authority (CS TA) Positioning, Navigation, and Timing (PNT) Reference Architecture (RA)

(Copies can be found at NAVWAR Enterprise Architecture (EA) - SYSCOMTA-ARC-023-PNTRA-V5.0 (CUI).pdf - AGB (sharepoint-mil.us)

MIL-STD-1399	Interface Standard for Shipboard Systems

(Copies can be found at ASSIST-QuickSearch Basic Search (dla.mil))

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Specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. DEFINITIONS

3.1 Time. Time signifies epoch, that is, the designation of an instant on a selected time scale, astronomical or atomic. It is used-in the sense of time of day.

3.2 Time interval. Time interval indicates the duration of a segment of time without reference to when the time interval begins or ends. Time interval may be expressed in seconds of time.

3.3 Precise frequency. Precise frequency signifies a frequency requirement to within one part in 10^9 of an established time scale.

3.4 Precise time. Precise time signifies a time requirement within ten milliseconds (ms).

4. GENERAL REQUIREMENTS

4.1 Requirements. The specific interface requirements and constraints established hereby are mandatory and shall be adhered to by the Department, of the Navy and all others engaged in any aspect of shipboard equipment design, production, and installation wherein that equipment is intended to interface with the PTTI PDS for support.

5. DETAILED REQUIREMENTS

5.1 General considerations. As weapons, navigation, command and control, communications, and other systems become increasingly sophisticated, the requirements for accurate, stable, non-interruptible time and frequency support also increase. The reference standards required to fulfill these requirements, for example, the 0-1695A/U Cesium Beam Frequency Standard and the AN/URQ-23 Frequency-Time Standard are expensive, and each user system having its own built-in standard is not cost effective. The PTTI program was established to provide Navy platforms and other users with PTTI reference standards traceable to the United States Naval Observatory (USNO) and to forestall thereby the proliferation of built-in standards.

5.1.1 PTTI system concept. The PTTI system consists of two major systems: The worldwide primary and backup dissemination system; and, the PDS. As used in this discussion, the term platform includes ships, submarines, aircraft, and shore installations. FIGURE 2 is a functional diagram showing the interrelationships between these major systems. The USNO has been tasked by the Department of Defense (DOD) to establish, coordinate, and maintain the standards for time and time interval for use by all DOD components, for example, Army, Navy, Air Force, National Security Agency, naval laboratories, and so forth. To provide accurate time for the DOD community, USNO maintains a master clock system which is referenced to the coordinated universal time (UTC) scale. USNO time thus referenced to UTC is identified as UTC (USNO).

Via the worldwide primary and backup dissemination system and the PDS, each equipment and system on board a Navy platform having a requirement for PTTI will be provided with time and time interval signals which will be directly traceable to the USNO, its master clock system, and UTC (USNO).

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The PTTI program will establish this capability of disseminating and distributing PTTI from the USNO on a continuous worldwide basis by using not only dedicated resources such as Navigation System Using Time and Ranging Global Positioning System (NAVSTAR GPS) , but also by using existing assets to the maximum extent practicable; for example, there are many communication and navigation systems presently in use within the DoD community which have an inherent capability to disseminate PTTI signals on a noninterference basis with the systems' primary missions. .

5.1.1.1 PTTI dissemination system. FIGURE 3 is a functional diagram showing an example of a PTTI dissemination system. (Reference CJCSI 6130.01G). The key point of this example is that precise time and time interval traceable to the USNO can be disseminated to various users over a variety of transmission systems. . It's also important to note that per CJCSI 6130.01G and OPNAVINST 9420.1C, the primary means of disseminating UTC (USNO) to platforms is via GPS Precise Positioning Service (PPS) which requires using keyed military GPS receivers. A firm ground rule of this program is that existing facilities must be used wherever practicable to meet current and planned user requirements. As future user system requirements become better defined and technological improvements in timing are realized, the dissemination system(s) will be upgraded.

5.1.1.2 PDS. FIGURE 4 is an example of a distributed PTTI PDS. A distribution system of this type will nominally employ three local reference standards. The distributed aspect of the system refers to the physical isolation of the local standards from each other for the purposes of PTTI distribution system survivability and availability; that is, the vulnerability of the distribution system to fire or battle damage is diminished by the spatial separation, and a fallback capability will generally be available. In addition, the "distributed" concept will tend to result in reference standards being powered by several platform power distribution networks, again increasing system survivability. Another important aspect of the PTTI PDS shown in FIGURE 4 is the use of a PTTI dissemination system external to the platform. As stated in 5.1.1 and 5.1.1.1, there are currently numerous systems in operation which have an inherent capability to disseminate PTTI information on a non-interference basis with the systems' primary missions. The specific external dissemination systems to be utilized by a mobile platform must be determined by analysis of its mission, its PTTI requirement, and the capabilities or assets on board. In the case of a stationary platform, the same information, as well as its geographic location will be factors in evaluating which of the external dissemination systems should be utilized. Three primary benefits can derive from utilizing a PTTI PDS such as is shown in FIGURE 4. The first will be a reduction in total cost. Nominally, three standards will serve all users aboard, thus relieving the user equipment's and systems of the necessity for their own internal oscillators or stand-alone reference standards. This should also comply with Chief of Naval Operations (CNO) guidance for the PTTI program by reducing the number of reference standards per platform and thus reduce the proliferation of standards in the fleet. The second benefit is that al l user systems will be provided with more accurate time and time interval information with a higher level of integrity. This results from two features of the PDS. First, each standard aboard will be compared with al l other standards aboard and the external dissemination system, which increases the integrity of the external data and allows better fault detection of individual sensors. Secondly, all standards aboard will receive regular time and time interval updates traceable to the USNO via the external dissemination systems. In the event that the external dissemination systems are lost, the platform falls back to the inter-comparison of the standards on board. This provides primary, via the external dissemination system, and secondary, via the on board reference standards, sources of PTTI and aligns with policy and standards, like the PNT RA.

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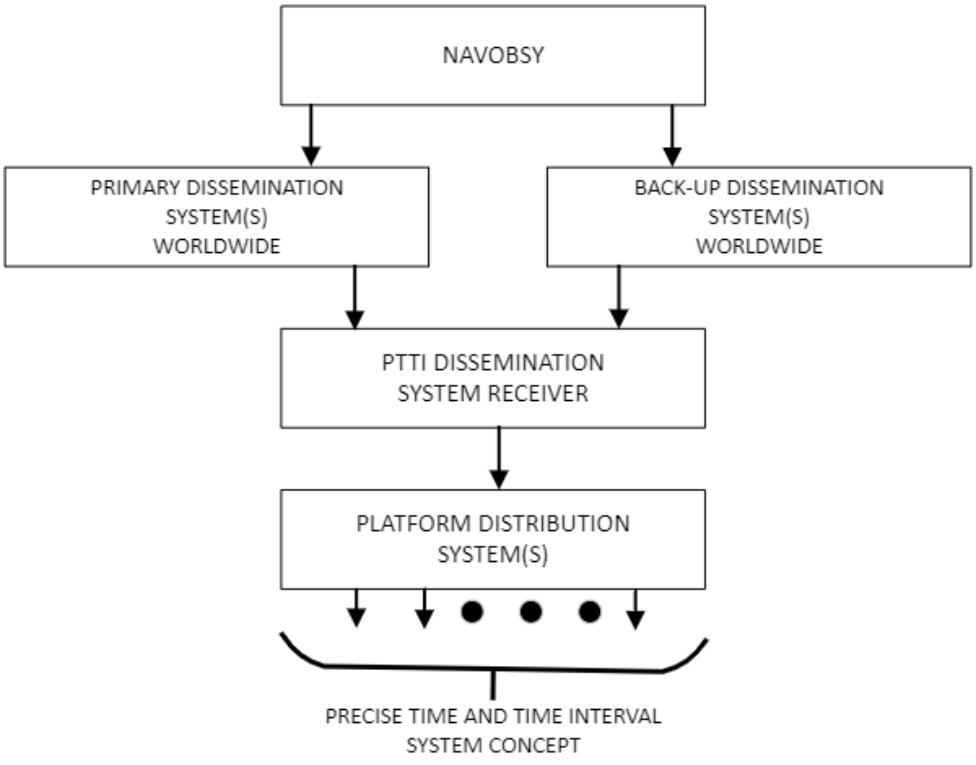


FIGURE 2. PTTI System concept.

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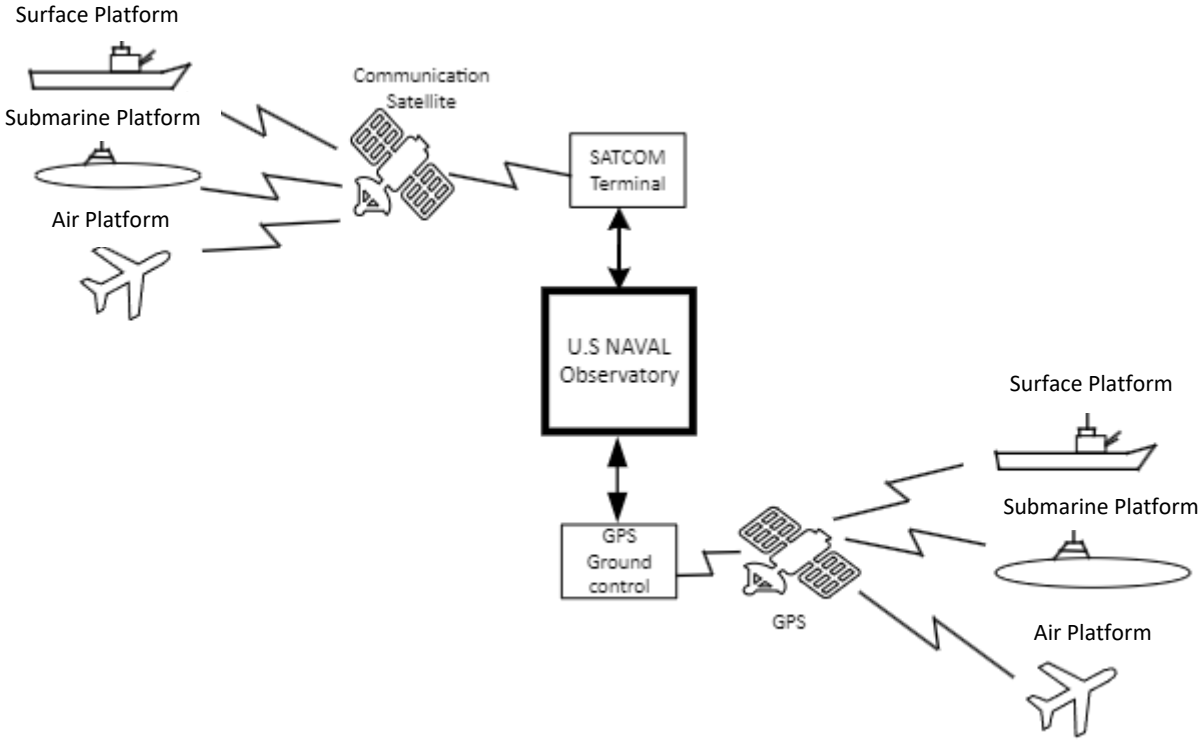


FIGURE 3. PTTI dissemination subsystem concept.

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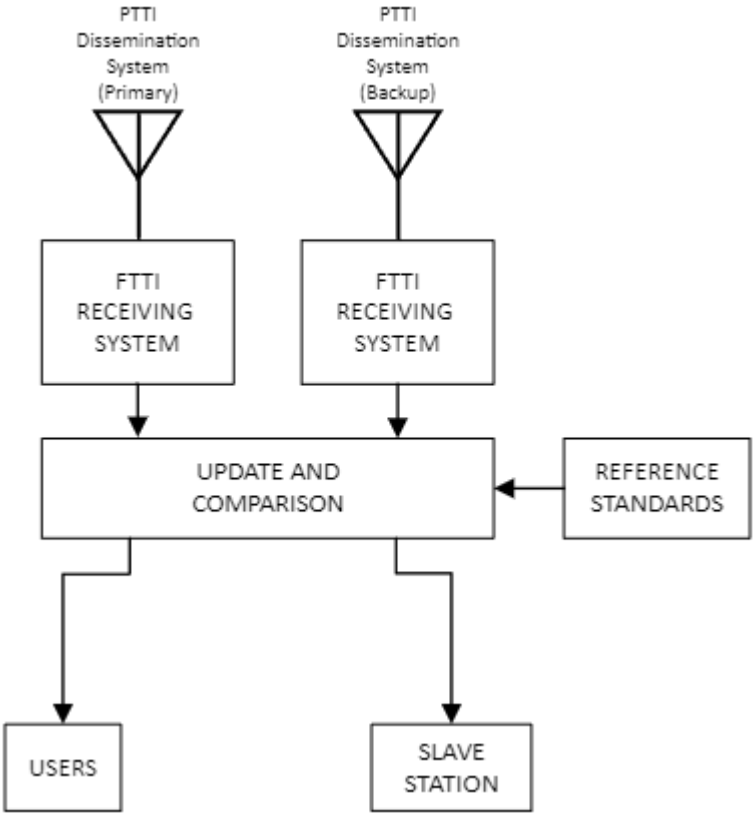


FIGURE 4. PTTI platform distribution system concept.

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5.2 Interface characteristics. The interface characteristics of the PTTI PDS are given below.

5.2.1 Frequency reference. Under normal conditions, the output frequencies provided will be 100 kilohertz (kHz), 1 megahertz (MHz), and 5 MHz with a long-term stability of five parts in 10 per day or better.

5.2.1.1 Harmonic distortion. Harmonic distortion levels will be at least 40 decibels (dB) below the required output levels.

5.2.1.2 Non-harmonically-related output. The level of any signal component not harmonically related to the output reference will be at least 80 dB below the required output level.

5.2.1.3 Frequency reference output voltage. The voltage at the output of the PTTI PDS for the frequency references of 5.2.1 will be at +1.0 (+0.5, -0.2) volts root-mean-square (V rms) into 50 ohms.

5.2.2 Time reference. For Have Quick time reference: refer to ICD-GPS-060A. For Binary Code Decimal (BCD) and Pulse per Second (PPS) refer to ICD-GPS-060B. For Inter-Range Instrumentation Group (IRIG) Serial Time Code Formats IRIG Standard 200-98, Network Timing Protocol (NTP) reference: IETF 1305, and Precise Time Protocol (PTP): reference IEEE 15288-2008. For surface platform specific tailoring of IRIG-B control functions refer to GPNTS documentation.

5.2.3 Interface between functional areas. The interfaces between equipment areas of the PTTI PDS will be as described in 5.2 except as specified below.

5.2.4 Buffering. Each output from the PDS will be individually buffered.

5.2.5 PTTI PDS capacity. PDS capacity is dependent on PTTI and PDS architecture and should be flexible enough to expand to accommodate new users.

5.3 Interface constraints.

5.3.1 Compatibility. The design of equipment intended to utilize the PTTI PDS shall be compatible with the interface characteristics described in 5.2.

6. NOTES

6.1 Conditions. In achieving the purpose of this section, it is recognized that there must be some flexibility of application. During the early design stage of shipboard equipment intended to utilize the PTTI PDS it may become apparent that significant advantages in the overall design or operation of such equipment can be achieved by deviating from the standard characteristics specified herein. In such instance, the provisions must comply with the Deviations of DoD-STD-1399A sect 441.

6.1.1 Deviation procedure. Requests for deviations will be submitted to NAVWAR with copies to:

a. Program or Project Manager

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b. Naval Sea Systems Command

c. PEO-C4I Assured GPS and PNT

6.1.2 Subject term (key word) listing

Coordinated Universal Time (UTC)
Global Positioning Navigation and Timing System (GPNTS)
Global Positioning System (GPS)
Integrated Radio Communications System (iRCS)
Naval Critical Distribution System (NCDS)
Navigation System Using Time and Ranging Global Positioning System (NAVSTAR GPS)
Platform Distribution System (PDS)
Positioning, Navigation and Timing (PNT)
Time Distribution Units (TDUs)
Time and Frequency Distribution System (TFDS)

6.1.3 Changes from previous issue.

Updated Section 2.1 – List of documents
Updated Section 5.1 – General considerations
Updated Section 5.1.1 – PTTI system concept
Updated Section 5.1.1.2 – PDS
Updated Figure 3 – PTTI dissemination subsystem concept
Updated Figure 4 – PTTI platform distribution system concept
Updated Section 5.2 – Interface characteristics
Updated Section 5.2.1 – Frequency reference
Updated Section 5.2.2 – Time reference
Removed Table I – Time references
Removed Figure 5 – BCD time code
Removed Figure 6 – 2137/2137 (DC) time codes
Removed Figure 7 – 1 PPS and I1 PPM timing signals
Removed Section 5.2.2.1 – Time reference output voltage
Removed Table II – Time reference output voltages
Removed Section 5.2.3.1 – Cesium beam frequency standard (0-1695A/U)
Removed Section 5.2.3.2 – Frequency standard (AN/URQ-23)
Updated Section 6.1.1 – Deviation procedure

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